Karlsruhe University of Applied Sciences

Department of Computer Science and Business Information Systems

Business Information Systems

BACHELOR´S THESIS

Efficiency vs. Performance in Green AI: An Empirical Investigation of Energy Consumption, Carbon Footprint, and Predictive Accuracy of Classification Algorithms at Varying Dataset Sizes.

Author	Eron Qorri
Matriculation number	85383
Workplace	Hochschule Karlsruhe
First Advisor	Prof. Dr. rer. nat. Christine Preisach
Second Advisor	Prof. Dr. Reimar Hofmann
Closing Date	31 July, 2026

31 July, 2026

Chairman of the Examination Board

Contents

Chapter 1

Introduction

1.1 Problem statement and societal relevance

The current trend in machine learning is moving towards increasingly large models and datasets [1]. Although deep learning approaches often achieve the highest predictive accuracy, their energy demand and associated CO_2 emissions increase disproportionately [2, 3]. In an era marked by strict sustainability guidelines and rising energy costs, a critical question arises: At what point is a computationally "simple" model ecologically more viable than a highly complex one?

The primary objective of this thesis is to determine whether and when it is worthwhile to deploy more complex models, considering not only their predictive accuracy but also their associated energy consumption. To achieve this, a Logistic Regression, a Random Forest, an XGBoost model and a Multilayer Perceptron are compared under equal conditions across three datasets of varying sizes. This comparative analysis aims to provide a clearer understanding of the thresholds at which simpler models represent the better choice. Additionally, the study investigates how modifying a Multilayer Perceptron architecture, specifically varying the number of neurons and hidden layers, directly impacts energy consumption and CO_2 emissions.

To address the problem statement and achieve the outlined objectives, this thesis investigates the overarching question: *How does the relationship between energy consumption and predictive accuracy change for Logistic Regression, Random Forest, XGBoost, and Multilayer Perceptron when model complexity and dataset size are systematically varied?*

Specifically, the following sub-questions are examined:

- How do energy consumption (in kWh) and CO_2 emissions change when scaling a deep learning architecture (variation of neuron count)?
- Under which conditions (dataset size, model complexity) does the resource consumption of complex models exceed the achievable accuracy gain compared to simpler algorithms?
- How does the energy consumption differ between CPU-based training (Random Forest) and GPU-accelerated training (XGBoost, MLP) for the same classification task?

1.2 Motivation: When does a simpler model pay off?

1.3 Research questions and objectives

1.4 Structure of the Thesis

The remainder of this thesis is structured as follows: Chapter 2 provides the theoretical background, distinguishing between Green AI and Red AI, and introducing the relevant sustainability and classification metrics. Chapter 3 details the methodology, including dataset selection and the experimental hardware setup. Chapter 4 outlines the implementation process, focusing on data preprocessing, model training, and the logging of resource consumption. The empirical results are analyzed in Chapter 5, evaluating both performance and resource utilization across scaling dataset sizes. Chapter 6 discusses the findings, highlighting the economic and environmental implications. Finally, Chapter 7 concludes the thesis and provides actionable recommendations for sustainable machine learning practices.

$$S = \frac{F_1}{1 + \lambda_1 \cdot t_{\text{scaled}} + \lambda_2 \cdot c_{\text{scaled}}}$$

Chapter 2

Theoretical Background

Chapter 3

Related Work

Table 1: Data Set Summary

Classification Data Set	Code	# Data Points (N_P)	# Features (N_F)
Annealing [20]	ANN	798	38
Breast Cancer [21]	BCW	683	9
Bank Marketing [22]	BMK	41188	20
Cardiotocography [23]	CAR	2126	21
Car Evaluation [24]	CEV	1728	6
Congressional Voting Records [25]	CVR	435	16
Credit Card Defaults [26]	DCC	30000	23
Diabetic Retinopathy [27]	DRB	1151	19
Handwritten Digit Recognition [28]	DRE	5619	64
Indian Liver Patient [29]	ILP	583	10
Ionosphere [30]	ION	351	34
Letter Recognition [31]	LRE	20000	16
Gamma Telescope Particles [32]	MAG	19020	10
Mice Protein Expression [33]	MPE	972	80
Mushroom Edibility [34]	MSH	8124	22
Occupancy Detection [35]	OCD	20560	5
Phishing Websites [36]	PHW	11044	30
QSAR Biodegradability [37]	QBD	1055	41
Skin Segmentation [38]	SSG	245057	3
Wall-Following Robots [39]	WFR	5456	24

I could do something of that sort for my own data sets

We also gathered several other characteristics of each data set, including number of categorical features (NCAT), number of continuous features (NCONT), proportion of features that are categorical (RCAT), number of target classes (NC), minimum single-class proportion (CMIN), and class imbalance (IC). The latter is a measure of the difference in actual and expected proportion of the least common target class (L) and its expected value (LE); i.e., I = (LE – L)/LE. These values are shown in Table 2.

Table 2: Additional Data Set Characteristics

Set	N_{CAT}	N_{CONT}	R_{CAT}	N_C	C_{MIN}	I_C
ANN	32	6	0.842	5	0.001	0.956
BCW	0	9	0.000	2	0.350	0.300
BMK	10	10	0.500	2	0.113	0.775
CAR	0	21	0.000	3	0.083	0.752
CVE	6	0	1.000	4	0.038	0.850
CVR	16	0	1.000	2	0.386	0.228
DCC	3	20	0.130	2	0.221	0.558
DRB	0	19	0.000	2	0.469	0.062
DRE	0	64	0.000	10	0.098	0.016
ILP	1	9	0.100	2	0.286	0.427
ION	0	34	0.000	2	0.359	0.282
LRE	0	16	0.000	26	0.037	0.046
MAG	0	10	0.000	2	0.322	0.357
MPE	3	77	0.038	8	0.092	0.259
MSH	22	0	1.000	2	0.482	0.036
OCD	0	5	0.000	2	0.231	0.538
PHW	30	0	1.000	2	0.443	0.114
QBD	0	41	0.000	2	0.337	0.325
SSG	0	3	0.000	2	0.208	0.585
WFR	0	24	0.000	4	0.060	0.760

During the MLP training experiments, several configurations were evaluated to optimize performance. Initially, the model was trained using only the CPU, which proved too slow for practical use. Switching to CUDA-based GPU training introduced a new issue: the default batch size of 128 provided by `scikit-learn` resulted in training times even slower than on CPU, due to the overhead of frequent small data transfers to the GPU.

Increasing the batch size to 8,192 significantly reduced training time. Further tuning of this parameter did not yield improvements, so this value was retained. Additionally, `pin_memory=True` was evaluated, which locks data in RAM to accelerate CPU-to-GPU transfers. However, this option also led to slower training, likely because the MLP architecture is not complex enough to benefit from the reduced transfer overhead. These experiments led to the final configuration used for all reported MLP results.

Chapter 4

Experimental Design

This chapter outlines the experimental design used to evaluate the ecological efficiency of different machine learning classification algorithms. It describes the datasets, models, evaluation metrics, hyperparameter tuning and measurement setup that form the basis of this empirical analysis.

4.1 Research Questions

The main research question that this study wants to find the answer to, is: *How does the trade-off between energy consumption and predictive performance vary across Random Forest, XGBoost, and MLP when model complexity and dataset size are systematically varied?* Further the following Sub-questions will be answered in the progress of this thesis:

- Under what conditions do the resource consumption of more complex models exceed the achievable accuracy gain over simpler baselines?
- How does energy consumption and CO2 output scale with increasing neural network complexity (number of layers and neurons)?
- How does energy consumption differ between CPU-based and GPU-based training for equivalent tasks?
- How does reducing the number of training instances on large datasets affect model accuracy and energy consumption across different algorithms?

To answer these questions multiple classification algorithms and datasets were used in the experimental phase of this empirical analysis.

4.2 Datasets

The following three datasets were selected, ranging from small to large scale, to assess how dataset size affects the relationship between model accuracy, energy consumption and carbon footprint.

The *Wine* dataset contains 178 instances across 13 features, where each feature describes a chemical property of wines from the same region in Italy, derived from three different cultivars.

Table 4.1: Overview of datasets used in this analysis

Dataset	**Instances**	**Features**	**Classes**	**Scale**
Wine [4]	178	13	3	Small
Credit Card Clients [5]	30,000	23	2	Medium
HIGGS [6]	11,000,000	28	2	Large

The *Default of Credit Card Clients* dataset contains 30,000 instances across 23 features and focuses on predicting payment default among credit card clients in Taiwan.

The *HIGGS* dataset is the largest dataset used in this study, comprising 11,000,000 instances across 28 features, seven of which are high-level features derived by physicists to help discriminate between the two classes [7]. The classification goal is to distinguish between signal events produced by Higgs bosons and background noise processes.

Collectively, these three datasets span several orders of magnitude in size, thus enabling a systematic evaluation of how model performance and energy consumption scale across varying levels of data complexity.

4.3 Models

For this empirical analysis multiple models were used to provide thorough answers for the research questions mentioned above. They range from baseline to Tree-based models to Neural Networks.

Table 4.2: Overview of classification models used in this study

Model	**Type**	**Device**	**Role**
Logistic Regression	Linear	CPU	Baseline
Random Forest	Ensemble	CPU	Mid-complexity
XGBoost	Gradient Boosting	CPU	Mid-complexity
XGBoost	Gradient Boosting	GPU	High-complexity
MLP	Neural Network	GPU	High-complexity

Bibliography

[1] R. Schwartz, J. Dodge, N. A. Smith, and O. Etzioni, "Green AI," *Commun. ACM*, vol. 63, no. 12, pp. 54–63, Nov. 2020, ISSN: 0001-0782. DOI: 10.1145/3381831 Accessed: Mar. 19, 2026.

[2] E. Strubell, A. Ganesh, and A. McCallum, *Energy and Policy Considerations for Deep Learning in NLP*, Jun. 2019. DOI: 10.48550/arXiv.1906.02243 arXiv: 1906.02243 [cs]. Accessed: Mar. 19, 2026.

[3] A. Lacoste, A. Luccioni, V. Schmidt, and T. Dandres, *Quantifying the Carbon Emissions of Machine Learning*, Nov. 2019. DOI: 10.48550/arXiv.1910.09700 arXiv: 1910.09700 [cs]. Accessed: Mar. 19, 2026.

[4] M. F. Stefan Aeberhard, *Wine*, 1992. DOI: 10.24432/C5PC7J Accessed: Mar. 19, 2026.

[5] I-Cheng Yeh, *Default of Credit Card Clients*, 2009. DOI: 10.24432/C55S3H Accessed: Mar. 19, 2026.

[6] P. Baldi, P. Sadowski, and D. Whiteson, "Searching for Exotic Particles in High-Energy Physics with Deep Learning," *Nature Communications*, vol. 5, no. 1, p. 4308, Jul. 2014, ISSN: 2041-1723. DOI: 10.1038/ncomms5308 arXiv: 1402.4735 [hep-ph]. Accessed: Mar. 19, 2026.

[7] D. Whiteson, *HIGGS*, 2014. DOI: 10.24432/C5V312 Accessed: Mar. 19, 2026.

List of Figures

List of Tables